Automated Water Quality & Contamination Detection Using Computer Vision, Machine Learning, Deep Learning and Generative AI

INDIVIDUAL THESIS

D. DODANWELA

IT22212122

Individual component: Microorganism (Bacteria) Detection

Group ID: R26-IT-005

Report submitted in partial fulfilment of the requirements for the

Bachelor of Science Honours in Information Technology

Specialized in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

October 2026 — Research design draft

DECLARATION AND SUPERVISOR CERTIFICATION

Candidate declaration. I declare that this thesis is my own work and does not incorporate, without acknowledgement, material previously submitted for a degree or diploma at any university or institute of higher learning. To the best of my knowledge, it contains no material previously published or written by another person except where acknowledgement is made in the text. I grant the Sri Lanka Institute of Information Technology the non-exclusive right to reproduce and distribute this thesis, in whole or in part, in print, electronic, or other media, while retaining the right to use the content in future work.

Academic integrity and AI disclosure. Generative AI assisted with drafting and structuring this document, as described in Appendix B. The candidate must verify the content and comply with the applicable module policy before signing the declaration.

Student name: D. DODANWELA

Student ID: IT22212122

Signature:  Date: 2026/10/11

Supervisor certification. The above candidate carried out this research for the bachelor's degree thesis under my supervision.

Role and name	Signature	Date
Supervisor: Mr. Buddika Harshanath		
Co-supervisor: Prof. Samantha Rajapaksha		

ABSTRACT

Automated analysis of water-sample microscopy images offers a potential means of supporting microorganism screening through spatial localization and evidence-supported identification. However, morphological similarities, small object sizes, overlapping cells and variations in image acquisition present challenges for reliable detection and taxonomic classification. This study proposes a deep-learning-based Microorganism Detection component within an integrated water-quality assessment system.

The proposed methodology involves establishing a defined microscopy protocol, curating reference-labelled images, developing instance-level annotations and comparing a YOLO-family detector with Faster R-CNN. Dataset partitions will be separated by independent specimens or preparation to minimize data leakage. The component will generate bounding boxes, predicted taxonomic labels, confidence scores and explicit identification limitations. Species- or genus-level outputs will be restricted to categories supported by reference evidence and validation, while insufficiently supported predictions will remain unidentified. Evaluation will distinguish localization performance from identification accuracy and will examine class-wise average precision, small-object recall, rejection of unfamiliar organisms, prediction coverage, robustness to imaging conditions and inference latency.

The anticipated contribution is a reproducible microscopy-analysis pipeline incorporating a traceable dataset, an evidence-based taxonomic reporting policy and a structured interface to the Water Safety Report and Purification Advisor. Model development and experimental evaluation remain pending; therefore, no empirical performance claims are presented. The proposed component is intended to support supervised research screening, and its predictions will not independently establish microorganism viability, pathogenicity, concentration or drinking-water safety.

Keywords: water-quality assessment, microscopy, microorganism detection, deep learning, object detection, taxonomic identification.

ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to my supervisor, Mr. Buddika Harshanath, and my co supervisor, Prof. Samantha Rajapaksha, for their guidance, constructive feedback and encouragement throughout this research. I also thank the Department of Information Technology at the Sri Lanka Institute of Information Technology (SLIIT) for the academic support and opportunity to undertake this study. My appreciation extends to my research group members for their collaboration, shared ideas and support in developing the overall water-quality assessment project. Their contributions have helped shape my individual component, the Water Safety Report and Purification Advisor. Finally, I am grateful to my family and friends for their patience, understanding and continued encouragement throughout my academic journey.

CONTENTS

1. INTRODUCTION 1

1.1 Overall project context 1

1.2 Individual component context 1

..... 2

1.3 Problem statement 2

1.4 Research gap 3

1.5 Aim and objectives 3

1.6 Research questions or hypotheses 3

1.7 Scope limitations and assumptions 4

1.8 Individual contribution 4

1.9 Thesis structure 4

2. LITERATURE REVIEW AND RELATED WORK 6

2.1 Review method and scope 6

2.2 Domain and theoretical foundations 6

2.3 Related research 7

2.4 Existing systems products and techniques 8

2.5 Gap synthesis and conceptual framework 8

3. RESEARCH METHODOLOGY 9

3.1 Research design 9

3.2 Data participants or study materials 9

3.3 Experimental or study procedure 12

3.4 Measures and evaluation metrics 12

3.5 Data analysis 13

3.6 Ethics privacy security and responsible AI 14

3.7 Reproducibility and research artefacts 14

4. PRODUCT AND SOLUTION DESIGN 15

4.1 Requirements 15
4.2 Component architecture within the complete solution 15
4.3 Design alternatives and decisions 16
4.4 Detailed design 16
4.5 Standards and professional practice 17
5. IMPLEMENTATION INTEGRATION AND TESTING 18
5.1 Development environment and technology stack 18
5.2 Component implementation 18
5.3 Integration with the other components 18
5.4 Verification and software testing 18
5.5 Deployment and operation 19
6. RESULTS AND VALIDATION 20
6.1 Research results 20
6.2 Product and technical results 20
6.3 User or stakeholder validation 21
6.4 Integration results 21
6.5 Summary of findings 21
7. DISCUSSION 22
7.1 Interpretation of findings 22
7.2 Comparison with literature baselines and existing solutions 22
7.3 Research contribution 22
7.4 Product contribution 22
7.5 Limitations and threats to validity 23
7.6 Implications 23
8. CONCLUSION AND FUTURE DIRECTIONS 24
8.1 Conclusion 24
8.2 Achievement of objectives 24

8.3 Individual research and product contributions 24
8.4 Future directions 25
REFERENCES 26
GLOSSARY 28
APPENDICES 31

LIST OF FIGURES

Figure 1: The YOLO Detection System 2
Figure 2: Faster R-CNN 2
Figure 3: Recall vs. IoU overlap ratio on the PASCAL VOC 2007 test set 6
Figure 4: Overview of the DL tasks and datasets used in DeepBacs 10
Figure 5: Segmentation of bacterial images using open-source deep learning approaches 10
Figure 6: DL-based object detection and classification 11
Figure 7: mage denoising for improved live-cell imaging in bacteriology 11

LIST OF TABLES

Table 1: Related work and applicability 7
Table 2: Reference and annotation policy 11
Table 3: Evaluation measures 13
Table 4: Output contract 16
Table 5: Results reporting matrix 20
Table 6: Objective status 24
Table 7: Image Acquisition Record 31
Table 8: Proposed Reporting Rules 32

LIST OF ABBREVIATIONS

The abbreviations used in this thesis are listed below.

Abbreviation	Meaning
AP	Average precision
API	Application programming interface
CNN	Convolutional neural network
DIBaS	Digital Images of Bacterial Species dataset
GPU	Graphics processing unit
IoU	Intersection over union
JSON	JavaScript Object Notation
mAP	Mean average precision
NMS	Non-maximum suppression
R-CNN	Region-based convolutional neural network
ROI	Region of interest
YOLO	You Only Look Once

1. INTRODUCTION

1.1 Overall project context

The overall research project proposes an integrated water-quality assessment system comprising photographic contamination detection, physicochemical water-quality classification, microorganism detection through microscopy, and automated report generation with purification guidance. These complementary components combine visual observations, measured parameters and microscopic findings to support a structured interpretation of water-quality conditions.

This thesis focuses on the Microorganism Detection component, which will analyze water-sample microscopy images to localize microorganisms and generate taxonomic predictions at a level supported by reference data and validation. Its output will include bounding boxes, predicted labels, confidence scores and identification limitations. These findings will be transferred to the reporting component while preserving their provenance and uncertainty. Microscopy-based predictions will remain distinguishable from physicochemical measurements and photographic observations, supporting transparent interpretation within the integrated system.

1.2 Individual component context

The proposed Microorganism Detection component will analyze microscopy images associated with individual water samples to localize microorganisms and assign supported organism labels. A convolutional-neural-network-based detector from the YOLO family [1] will serve as the primary candidate, with Faster R-CNN [2] providing a comparative architecture. Outputs will comprise bounding-box coordinates, predicted labels, confidence scores and image-quality information, enabling traceable interpretation within the integrated water-quality assessment system.

Taxonomic predictions will be restricted to species or genus levels where reference annotations, the selected imaging modality and experimental validation support such distinctions. Where evidence is insufficient, the component will return an unidentified or insufficient-evidence outcome. Any optional object counts will represent detected instances within the analyzed field of view and will not be interpreted as microbial concentrations in the original water sample without a separately validated quantification procedure.

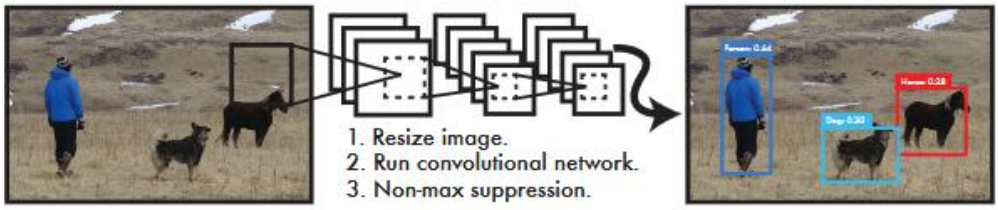


Figure 1: The YOLO Detection System

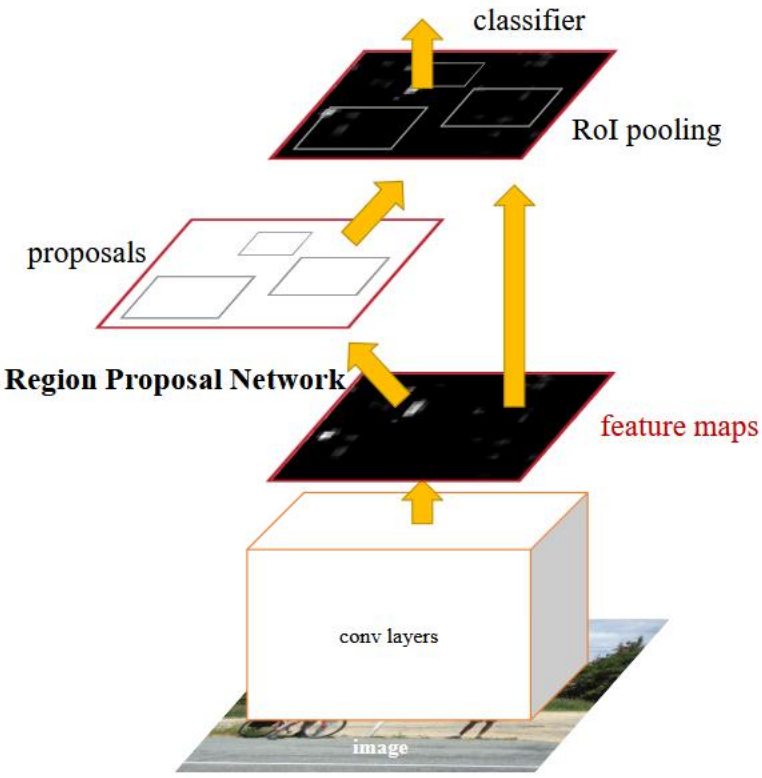


Figure 2: Faster R-CNN

1.3 Problem statement

Microscope images can contain small, overlapping and visually similar cells alongside non-biological particles or preparation artefacts. A useful automated method must distinguish detection of a cell-like object from evidence for its taxonomic identity. An image-level label alone also fails to show which individual region supports a prediction.

The research problem is to develop and evaluate a detector that localizes supported microorganism categories while avoiding unsupported species assignments. It must preserve sample provenance, cope with imaging variability and report uncertainty in a form the downstream advisor can interpret.

1.4 Research gap

General detection architecture provides localization mechanisms [1], [2] and microscopy studies demonstrate useful deep-learning applications [3]-[5]. However, results obtained with preparations, taxa and microscopes do not establish accuracy on mixed environmental water samples. The project therefore requires task-specific reference labels, specimen-level evaluation and explicit limits on the taxonomic claims it will make.

The working gap is an evaluated integration of microorganism localization, supported genus/species prediction and uncertainty reporting within this group project. No claim that the architecture is novel or that no comparable system exists is made. A wider domain review and completed experiments are needed to establish the final research contribution.

1.5 Aim and objectives

Aim: To design and evaluate a CNN-based object-detection component that localizes microorganisms in water-sample microscopy images and predicts only the taxonomic labels supported by reference data and validation.

Objectives:

01: Define the imaging domain, target taxa, reference-identification requirements and annotation policy.

02: Curate a traceable microscopy dataset with instance annotations and independent development/test partitions.

03: Train and compare specified YOLO and Faster R-CNN detectors for localization and identification.

04: Evaluate uncertainty handling, genus-level fallback and rejection of unsupported taxonomic assignments.

05: Integrate the component with the group reporting system and measure robustness, latency and software reliability.

1.6 Research questions or hypotheses

Q1: How accurately do the candidate detectors localize target microorganisms on independent microscope preparations?

Q2: Which selected taxa can be discriminated against at species level, and which require genus-level or unidentified reporting?

Q3: How do object size, overlap, focus and microscope changes affect detection and identification?

Q4: Can validation-selected confidence and rejection rules reduce incorrect species assignments while retaining useful prediction coverage?

Q5: What localization, identification and latency trade-offs distinguish the candidate detector configurations?

The expectation that transfers learning and microscopy-specific annotations will improve performance is a hypothesis to test. No accuracy level or model superiority is assumed.

1.7 Scope limitations and assumptions

The initial study will use one documented microscopy modality and acquisition protocol. A feasible default is reference-labelled bright-field imagery, subject to availability and reviewer agreement. Phase-contrast, fluorescence and three-dimensional phase images will not be pooled without recording modality and evaluating compatibility. Magnification alone is insufficient metadata; calibration, resolution and preparation method also matter.

The initial taxonomic panel will be a small, feasible set selected after auditing data and laboratory support. E. coli is an example proposed in the group documents, not a promised capability. Coliform is not a single genus or species; cyanobacteria and protozoa are also broad groups and must not be mixed with species labels as if they were equivalent ranks.

The component excludes clinical diagnosis, strain typing, pathogen confirmation, viability assessment and autonomous water-safety decisions. It will not infer concentration per millilitre from a few image boxes. Environmental water performance must be evaluated separately if development data originates from laboratory reference preparations.

1.8 Individual contribution

The planned individual work includes the microscopy data specification, taxonomic ontology, reference-label register, box-annotation guide, dataset splits, training and model comparison, uncertainty policy, inference endpoint and evaluation artefacts. The candidate will also define how image-level results are passed to the group report generator.

Actual ownership should later be evidenced by code history, annotation records and experiment logs. The other components' numerical models, ordinary-image detectors and report-generation logic remain separate team contributions.

1.9 Thesis structure

Chapter 2 reviews detector foundations and relevant microscopy research. Chapter 3 describes the dataset, reference standard and evaluation. Chapter 4 presents the architecture and taxonomic output

policy. Chapter 5 outlines implementation, integration and testing. Chapter 6 records pending results, and Chapter 7 discusses interpretation and limitations. Chapter 8 concludes with objective status and future directions. The appendices provide review instruments and an AI-use disclosure.

2. LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

This draft uses a focused review of the supplied project descriptions, primary YOLO [1] and Faster R-CNN [2] papers, the DIBaS study [3], DeepBacs [4] and a species-identification study using three-dimensional quantitative phase imaging [5]. Source records were checked on 6 October 2026. The review considers the distinction between image classification, instance detection, segmentation and taxonomic identification.

This is not a systematic literature review. The final thesis should document additional searches on water-sample microscopy, mixed-species detection, small-cell localization and reference-standard validation. For each study, compare imaging modality, target taxa, preparation, annotation granularity, split unit and metric before judging its applicability.

2.2 Domain and theoretical foundations

A convolutional neural network (CNN) learns image features [6]; an object detector combines localization with category prediction. A bounding box indicates an image region, while a taxonomic label expresses an identity claim. Their correctness must be evaluated separately and together.

Intersection over union (IoU) is the overlap area divided by the union area of predicted and reference boxes [1]. Very small objects make pixel-level coordinate errors consequential. The detector's confidence score is not automatically calibrated and is not evidence of pathogenicity. Classification at species, genus and morphology levels uses different label meanings and must preserve the associated rank [7].

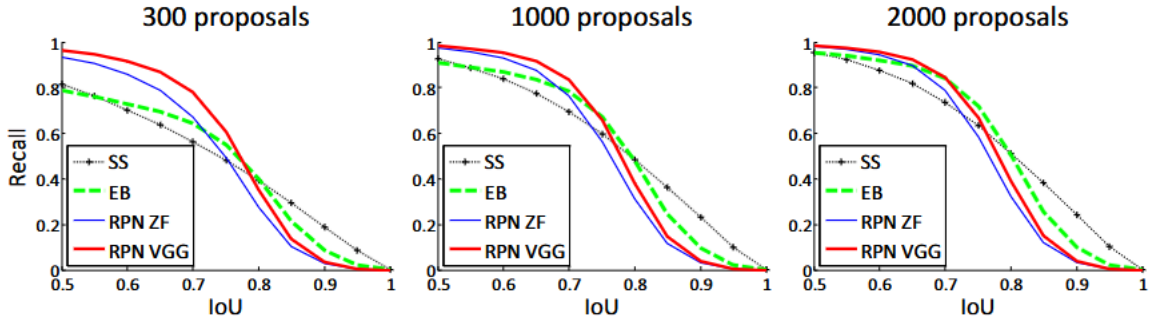


Figure 3: Recall vs. IoU overlap ratio on the PASCAL VOC 2007 test set

2.3 Related research

YOLO frames object detection as predicting boxes and class probabilities with one network evaluation [1]. Faster R-CNN uses a region proposal network sharing convolutional features with the detection network [2]. These establish candidate architectures, but their original benchmark outcomes do not transfer directly to microscopy.

Zieliński et al. [3] introduced DIBaS and a bacterial-image classification approach. The study reports 660 images spanning 33 genera/species and discusses morphological similarities that complicate identification. It is useful background, but image-level labels are not automatically verified boxes for individual organisms in mixed water images.

Spahn et al. [4] presented DeepBacs for bacterial image analysis, including segmentation and growth-stage object detection. Its tasks motivate microscopy-specific evaluation; a growth-stage label does not become a taxonomic label simply by changing a class name. Kim et al. [5] studied species identification using three-dimensional quantitative phase imaging. The specialized imaging conditions are central to interpreting that result and do not establish the same capability in routine two-dimensional bright-field images.

Moen et al. [6] review deep-learning methods for cellular image analysis, providing broader context for microscopy-specific data preparation and evaluation. Guo et al. [7] examine neural-network confidence calibration, motivating separate assessment of prediction confidence and correctness; their findings do not independently validate unknown-organism rejection in this project.

Ronneberger et al. [8] introduced U-Net for biomedical image segmentation, while Stringer et al. [9] developed Cellpose for cellular segmentation. Lugagne et al. [10] presented DeLTA for automated cell segmentation, tracking and lineage reconstruction. These studies inform segmentation and tracking alternatives, but their outputs do not by themselves establish species identity in mixed water samples.

Table 2.1 summarizes how these sources inform the proposed study. The project's claim is an evaluated application, not a new detector architecture.

Table 1: Related work and applicability

Source	Relevant contribution	Boundary for this project
[3] YOLO	Unified box and class prediction	General detector foundation; not water-specific validation.
[4] Faster R-CNN	Shared-feature region proposal detector	Comparison architecture; needs microscopy training.

Source	Relevant contribution	Boundary for this project
[5] DIBaS study	Bacterial-image classification dataset and method	Image labels do not provide mixed-field instance identity.
[6] DeepBacs	Bacterial segmentation and growth-stage detection	Task-specific labels must not be mistaken for species labels.
[7] 3D phase imaging	Species identification in specialized imagery	Different modality; cannot assume routine 2D performance.

2.4 Existing systems products and techniques

Possible approaches include manual microscopy review, whole-image classification, object detection, instance segmentation and specialized identification workflows. Manual review is necessary for annotation quality but must be supported by a suitable reference identity. Whole-image classification cannot establish which cell carries a label in a mixed field.

Object detection matches the requested bounding-box output. Instance segmentation may be useful for touching cells and precise boundaries but adds annotation requirements. Specialized imaging or independent laboratory identification may improve reference certainty but changes the acquisition scope and cost. The initial study will choose one supported domain before exploring additional modalities.

2.5 Gap synthesis and conceptual framework

The conceptual framework links reference-label quality and acquisition consistency to learned localization and identification. A quality gate checks whether the image falls within the intended domain. The detector generates candidate boxes; a validated label policy decides whether a supported species, genus or unidentified result may be returned. The output is then attached to the sample record.

Independent variables include detector configuration and selected preprocessing or tiling options. Outcomes include class-wise detection quality, taxonomic error, rejected-prediction coverage and latency. Microscope domain, specimen and cell density are factors for stratified analysis.

3. RESEARCH METHODOLOGY

3.1 Research design

A design-and-evaluate approach is proposed. The artefact will consist of the microscopy dataset specification, trained detection pipeline, uncertainty policy and integration service. Iterative development will use pilot annotations and a validation partition; the final evaluation will use independently held-out specimens or preparations.

Software verification and empirical model evaluation will remain distinct. Correct coordinate conversion establishes implementation behaviour, whereas species generalization requires independent reference-labelled data. The supplied documents describe a proposal and do not provide complete training or experiments.

3.2 Data participants or study materials

Candidate materials include permissioned laboratory microscopy and public research datasets. DIBaS [3] and DeepBacs [4] will first be audited for licensing, modality, available instance labels and suitability. They are candidate supporting resources rather than confirmed ready-to-use water-species detection datasets. Image-level labels must not be copied indiscriminately onto every particle in a field.

A planning target is approximately 1,000–2,000 fields of view across multiple independent specimens or preparations and a small panel of well-supported taxa. This is a feasibility target, not an existing collection or proof of sufficient statistical power. Final class counts depending on reference availability and annotation quality. Record specimens, slides, sessions, fields and object count separately.

Reference identity will be supplied by a qualified collaborating laboratory using an appropriate independently validated method, with its provenance retained. A reference-identified single-taxon preparation can support labels only after excluding debris and other artefacts. In mixed environmental samples, a sample-level positive result does not identify each imaged cell; unresolved instance identity must remain unknown. Table 3.1 defines the reference and annotation policy.

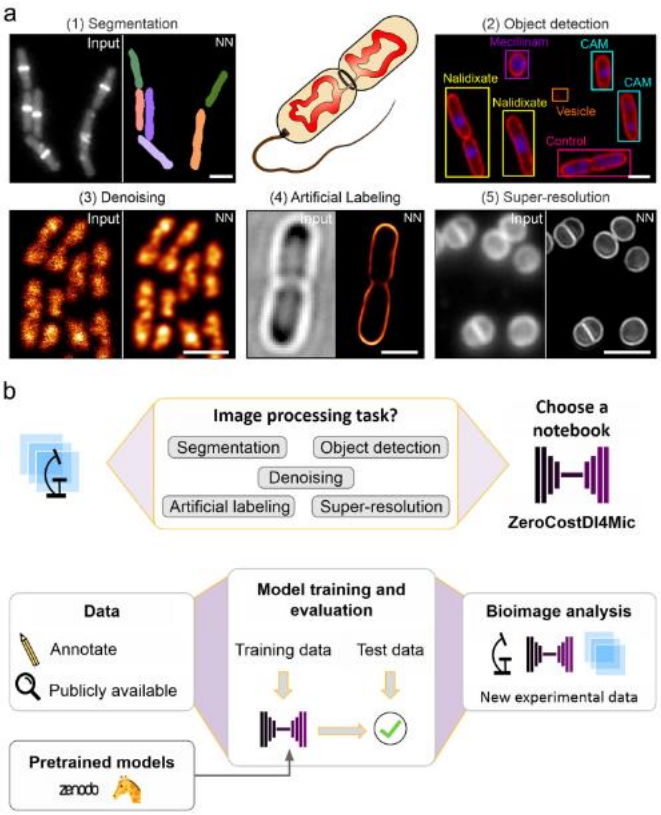


Figure 4: Overview of the DL tasks and datasets used in DeepBacs

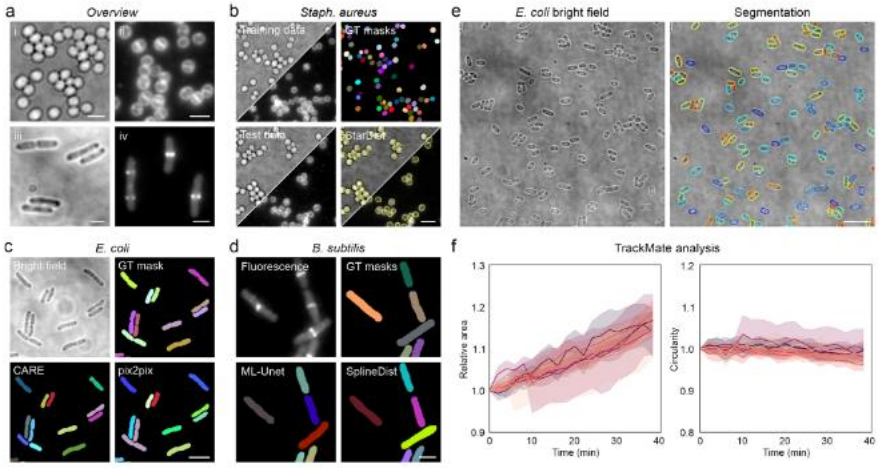


Figure 5: Segmentation of bacterial images using open-source deep learning approaches

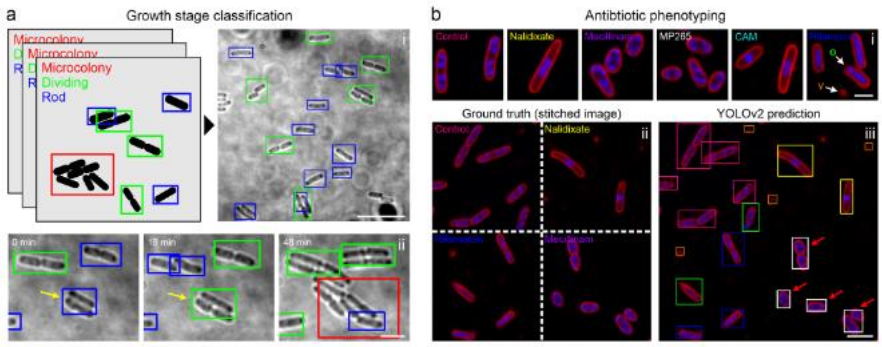


Figure 6: DL-based object detection and classification

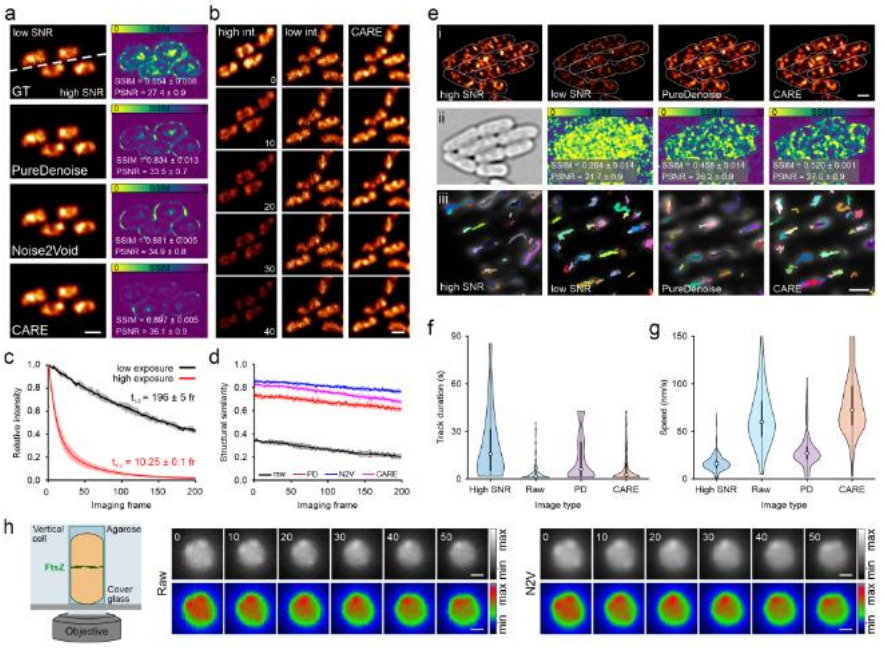


Figure 7: mage denoising for improved live-cell imaging in bacteriology

Table 2: Reference and annotation policy

Situation	**Required reference treatment**	**Output/annotation policy**
Reference-labelled single-taxon preparation	Document laboratory identity and preparation provenance	Box only verified cell-like instances; exclude artefacts.
Mixed environmental field	Require identity evidence at suitable instance granularity	Do not transfer a sample-level positive to every cell.
Species indistinguishable in selected domain	Review whether genus discrimination is supportable	Use validated genus label or unidentified.

Situation	Required reference treatment	Output/annotation policy
Dense unresolved cluster	Record cluster/ignore policy and count unit	No invented separate boxes or cell counts.
Non-target organism or artefact	Document known negative/unknown status	Use for rejection and false-positive evaluation.
Image-level public dataset	Audit licence, source and missing instance annotations	Supporting resources until suitable annotations exist.

3.3 Experimental or study procedure

First, audit available data and choose the microscopy modality and taxonomic panel. Record objective, numerical aperture when available, camera, pixel calibration, preparation/staining category and session. Pilot the annotation guide with two reviewers, retaining disagreements and revising definitions before large-scale annotation.

Assign independent specimens or preparations to approximately 70% training, 15% validation and 15% test partitions before generating crops or tiles. Keep every slide, field, tile, adjacent frame and augmented derivative from the same source group together. Where feasible, hold out strains or isolations within known taxa, as well as a separate acquisition session, to test whether predictions generalize beyond preparation-specific appearance.

Fine-tune a fully specified YOLO-family model and Faster R-CNN from documented checkpoints. Use identical data partitions, label definitions and evaluation code. Give both models comparable tuning budgets, while recording architecture-specific input sizes and optimizers. Choose confidence, tiling, suppression and stopping settings on validation data only. Repeat training with at least three seeds when resources allow.

Compare full-frame inference with overlapping tiling where small objects would otherwise be lost through resizing. Map tile predictions to the original field and merge duplicates before scoring. Preserve an independent set of non-target organisms and artefacts for rejection evaluation. Freeze the configuration before testing; use distinct held-out unknown examples for final testing where feasible.

3.4 Measures and evaluation metrics

Table 3.2 defines the planned evaluation. Species detection requires both a correct taxonomic label and a matched box. Genus-level evaluation will use an explicit species-to-genus mapping and will be reported separately. Predictions labelled unidentified can receive credit for class-agnostic localization but not for correct species identification.

Report average precision at IoU 0.50 and mAP across IoU thresholds from 0.50 to 0.95 in steps of 0.05, with evaluator configuration recorded. For tiny objects, present additional size-stratified and localization-error analysis rather than lowering the threshold after observing poor test results. Report negative-image errors and unsupported predictions, not only scores on images containing known targets.

Table 3: Evaluation measures

Metric	Reporting requirement
Class-agnostic localization	Correctly matched organism boxes independent of species label.
Species/genus AP and mAP	Correct box and label at stated rank; per-class and macro scores.
Precision / recall / F1	Validation-selected threshold; one-to-one class-aware matching.
Identity confusion	Matched-object species/genus errors plus end-to-end detection.
Unknown acceptance / known rejection	Both directions, with accepted-prediction accuracy and coverage.
Robustness	Size, density, focus, specimen and acquisition-domain strata.
Operational performance	Median/p95 latency, hardware, batch size, resolution and memory.

3.5 Data analysis

Compare models on the same test specimens. Report means and variability across seeds, and use specimen-level resampling for uncertainty intervals rather than treating thousands of cells from one slide as independent evidence. Stratify outcomes by taxon, size, overlap, focus and microscope session. Retain the full confusion matrix, including unidentified outputs.

Quantify the relationship between accepted-prediction accuracy and coverage as rejection becomes stricter. Report known-target false rejections and unknown-target false acceptances separately. Calculate conditional identity accuracy on matched objects and end-to-end species detection performance so missed cells are not excluded from the main conclusion.

Reviewer agreement should be reported before adjudication, with reasons for unresolved labels. Any counting analysis must state whether a reference unit is one cell, one cluster or one object; do not compare incompatible units.

3.6 Ethics privacy security and responsible AI

Prefer existing permissioned imagery or material generated by authorized collaborating laboratories. Collection, handling and preparation of microorganisms must remain within institutional approval and competent laboratory practice. This thesis provides a computational design, not organism-culture or pathogen-handling instructions.

Use de-identified sample codes, confirm image licences and restrict sensitive laboratory metadata. Annotation and user-review participation will follow institutional consent requirements. Reports must distinguish research predictions from confirmed identification. Upload validation, access control and audit logging will protect the service, while uncertain or unsupported outputs will remain visible rather than being silently forced into named taxa.

3.7 Reproducibility and research artefacts

Preserve dataset provenance, reference-identification evidence, taxonomic mapping, image calibration, annotation versions, split manifests, exclusion reasons, source code revision, dependency versions, random seeds and checkpoint hashes. Experiment records will include hyperparameters, validation selection, prediction files, matching settings, rejection thresholds and hardware timing.

Release only materials permitted by licences and agreements. Where raw images cannot be shared, provide allowed metadata, protocol descriptions and reconstruction instructions. The final model card must state its supported modality, taxa, known limitations and non-validated uses. Appendix A contains proposed review and test records.

4. PRODUCT AND SOLUTION DESIGN

4.1 Requirements

Functional requirements are microscope-image upload with specimen identity, metadata validation, quality screening, localized predictions, supported genus/species labels, confidence, explicit unknown outcomes, annotated previews and structured integration output. The service must distinguish no detected object, rejected image and inference failure.

Non-functional requirements include traceability, repeatable preprocessing, protected uploads, stable coordinate conventions, bounded resource use and measured latency. Species acceptance thresholds will be agreed using validation evidence and domain review. No unmeasured accuracy, real-time performance or clinical suitability is promised.

4.2 Component architecture within the complete solution

The offline path comprises reference-data acquisition, metadata review, annotation, grouped splitting, model training, validation and model registration. The online path comprises image validation, modality/quality checks, preprocessing, optional tiling, CNN detection, coordinate restoration, duplicate handling, taxonomic acceptance and result packaging.

The microscope-image interface provides sample, slide and field identifiers. The detector returns structured observations to the Water Safety Report and Purification Advisor, as specified in Table 4.1. The adviser should preserve identification rank, confidence and evidence limitations when combining these results with photographs and physicochemical measurements. The detector will not generate treatment advice.

Table 4: Output contract

Field	Meaning
sample_id / specimen_id / slide_id / image_id	Traceable identity at each acquisition level.
modality / calibration / dimensions	Recorded imaging domain and original pixel dimensions.
schema_version / model_version	Contract and exact detector/configuration revision.
status / quality_warnings	Detections, no detection, rejected input or inference failure.
detections[].box_xyxy	Original-image pixel coordinates with documented convention.
taxon_name / taxonomic_rank / confidence	Accepted species/genus or unidentified label and model score.
reference_scope / limitations	Research prediction domain; no implied laboratory confirmation.
object_count / count_unit	Optional field-level count, explicitly cell or cluster unit.
audit_id / latency_ms	Trace reference and measured processing time.

4.3 Design alternatives and decisions

YOLO and Faster R-CNN provide contrasting detector designs. The final choice will depend on supported-taxon detection, false assignments, small-object performance, latency and deployment constraints. Exact model variant, checkpoint, backbone and input configuration must be recorded.

Bounding boxes are appropriate for locating separable cells or defined organism instances. Dense unresolved clusters should not be divided into invented individual cells; they require the declared cluster or ignore-region policy. Segmentation remains a possible extension. A high confidence score alone will not justify species reporting where that species has not demonstrated reliable discrimination.

4.4 Detailed design

The label ontology will store class_id, taxon_name, taxonomic_rank and reference_basis. Species labels will map to a documented genus. Morphology labels such as rod-shaped cells will be kept separate from taxonomic names. The software will not relabel a broad group as a species or infer genius solely from an arbitrary shape category.

Every box will use original-image pixel coordinates with a documented axis and endpoint convention. Tiling will record tile offsets and resize transforms. Postprocessing will use validated settings to merge repeated observations of the same object without indiscriminately suppressing neighboring cells. Store both raw predictions and accepted outputs for audit.

The proposed acceptance policy combines image-domain eligibility, class-specific validation and confidence criteria. A species may be shown only if supported by the model's evaluated label set and the accepted prediction rule. Genus's fallback is allowed only when a separately evaluated genus decision is supported; it is not automatically justified by low species confidence. Otherwise return detected_object_unidentified or a qualified candidate label.

Closed-set confidence cannot reliably guarantee rejection of every unseen organism. Unknown and artefact sets will therefore be included in development and evaluation, and residual open-set limitations will be stated. If image metadata or quality is outside the supported domain, return a rejected or limited-result status.

Optional counts will refer to retained image objects or resolved cell instances under the annotation policy. They will not be reported as viable counts or concentration per volume. Aggregating overlapping fields or repeated time points require tracking and sampling information beyond this initial component.

4.5 Standards and professional practice

Professional practice will include a reviewed annotation manual, a reference-label register, dataset and model cards, reproducible experiment logs and versioned API contracts. Confirm software, pretrained weight and dataset licences before redistribution. Record the source of the taxonomic vocabulary and the edition or snapshot used.

Human-facing labels will preserve rank and uncertainty. The component does not claim laboratory accreditation, regulatory approval or diagnostic certification. A model update that changes supported taxa or confidence rules must trigger regression evaluation and a version change.

5. IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

A candidate environment uses Python, a deep-learning framework such as PyTorch, an image-processing library, an instance-annotation tool and a versioned web API. Training will use suitable GPU resources where available, while inference will be benchmarked on the actual target hardware. These are proposed choices, not a record of installed dependencies or completed training.

The final environment record should include operating system, framework and library versions, GPU and driver/runtime information, detector source revision and all model parameters needed to reproduce inference.

5.2 Component implementation

Implementation will begin with metadata and class-schema validation, followed by grouped dataset construction, annotation conversion, training configuration, model inference, coordinate mapping and output packaging. The uncertainty policy and image-quality gate will then be added and tested before integration.

Do not use interpolation or enhancement to claim newly resolved biological features. Any preprocessing must be consistent between development and use and evaluated for its effect on identity. Model weights and configuration will be loaded as a versioned unit. No implemented module or completed experiment was supplied, so this section remains a work specification.

5.3 Integration with the other components

The detection endpoint will accept sample_id, specimen_id, slide_id, image_id, modality metadata and the image. The response will contain original dimensions, quality status, model version, accepted detections and their identity rank, plus limitations and processing time. Mismatched identifiers or unsupported schemas will be rejected.

The reporting component must not transform an image-object count into organisms per millilitre or an uncertain prediction into a confirmed pathogen. Contract tests will verify that unknown labels and rejected-image states survive end-to-end. Integration will begin with mocked messages, followed by actual outputs from the group’s dashboard and report service.

5.4 Verification and software testing

Unit tests will cover file validation, metadata parsing, class-rank mapping, resize reversal, tile-offset arithmetic, boundary clipping and duplicate handling. Geometric fixtures will test coordinates

without depending on a trained model. Ontology tests will ensure that invalid species/genus mappings are rejected.

Model evaluation will use reference-labelled held-out imagery, known negatives and unfamiliar organisms or artefacts. Tests must distinguish detector misses from identification errors. Integration tests will cover empty outputs, missing metadata, out-of-domain images, service timeouts and malformed requests. Appendix A lists proposed cases; no executed test passes are claimed.

5.5 Deployment and operation

Deploy first in a restricted research environment with reviewed models and datasets. Protect sample metadata, impose upload and compute limits, and store secrets outside code. Keep model versions and rollback artefacts together. Agree retention periods for images and logs before deployment.

Monitors rejected inputs, processing failures, latency, unknown rates and reviewer-reported misidentifications. A new microscope, staining protocol or sample domain will require validation before claiming support. Human review should remain available for uncertain and consequential findings. Published benchmark speed is not an operational service guarantee.

6. RESULTS AND VALIDATION

6.1 Research results

No dataset statistics, trained checkpoints or experimental predictions were provided. Detection mAP, species/genus performance, uncertainty outcomes and model comparisons therefore remain pending. Table 6.1 identifies the required results.

The final chapter should begin with independent specimen counts, imaging conditions, label distributions and reference quality. Report realized dataset splits, exclusions and reviewer agreement before model metrics. State which taxa meet the predeclared evidence criteria, and which do not, without silently removing unsuccessful classes.

Table 5: Results reporting matrix

Evidence	Required final record	Current status
Dataset and reference quality	Specimen/image/object counts, taxa, modality, annotation agreement	Not provided
Localization and identity	Per-class AP, mAP, rank-specific errors and missed objects	Not evaluated
Rejection performance	Unknown acceptance, known rejection and coverage	Not evaluated
Robustness	Independent specimen and microscope-domain outcomes	Not evaluated
Software/integration	Coordinate checks, ontology tests and advisory trace	Not executed
Latency and resources	Measured runtime, resolution and hardware	Not measured

6.2 Product and technical results

Technical validation is pending. Required artefacts include the functioning inference service, representative annotated outputs, validated box restoration, tile-merging tests, ontology checks and runtime measurements. Report model-only time and full-request latency separately, with hardware and resolution.

An attractive microscopy overlay is not evidence of correct identity. Likewise, successful API calls do not establish generalization to environmental samples or unseen species. These conclusions require the corresponding labelled experiments.

6.3 User or stakeholder validation

No reviewer or stakeholder results are available. The proposed domain review will assess box quality, reference-label validity and appropriateness of species, genus and unidentified outputs. If a user study is conducted, ask participants to distinguish predictions from confirmation and interpret an unknown or no-detection result.

Recruitment, consent, study procedure and actual findings must be documented if performed. Do not invent approval, participant quotations or satisfaction scores. A small formative review can improve presentation but cannot validate taxonomic accuracy.

6.4 Integration results

End-to-end integration results remain unmeasured. Required evidence is a trace from one microscope image and specimen ID through the detector to the advisory report. Check that coordinate overlays align, class names retain rank, counts retain their image-level meaning and model provenance is preserved.

Demonstrate missing-image, rejected-image, uncertain-identification and timeout cases. The final report must not treat any of these as a negative laboratory finding.

6.5 Summary of findings

The current deliverable is a research design with a taxonomic label policy, annotation requirements, architecture and evaluation protocol. It does not establish successful microorganism identification or operational benefit. Performance and conclusions remain conditional on the experiments and reference evidence specified above.

7. DISCUSSION

7.1 Interpretation of findings

Future interpretation must distinguish localization success from identity success. Strong class-agnostic detection with weak species prediction would support screening for organism-like objects, not a species-identification claim. Good performance on reference preparations with poor environmental transfer would restrict the validated domain.

Improved accepted-prediction accuracy may simply reflect increased rejection; coverage must be reported alongside it. Dense-cell misses and false genus fallback require separate discussion. No empirical interpretation is offered here because results are not available.

7.2 Comparison with literature baselines and existing solutions

The architecture comparison builds on [1], [2], while [3]–[5] illustrate that microscopy tasks differ in image type, label granularity and available evidence. Whole-image bacterial classification, growth-stage detection and three-dimensional species identification cannot be treated as equivalent benchmarks for water-sample box detection.

Compare the project models on identical held-out specimens and disclose differences in compute budget. A detector with better general benchmark scores may still perform worse on the chosen microscopy domain. Literature results are context, not replacement evidence for this project.

7.3 Research contribution

The intended research contribution is a reproducible assessment of supported taxonomic identification together with localization under a defined microscopy protocol. The main artefacts are the instance-label policy, independent specimen split, rejection evaluation and error analysis. The contribution is application and evaluation work rather than invention of new CNN architecture.

Claims should be limited to taxa and acquisition conditions tested. Reference-label uncertainty and unresolved instances are research findings to report, not inconvenient cases to hide.

7.4 Product contribution

The planned product contribution is a traceable microscopy-analysis service that returns boxes and qualified organism identities to the group platform. Useful features include original-coordinate overlays, explicit taxonomic rank, unknown handling, model versioning and a stable interface.

Its practical value depends on demonstrated accuracy and appropriate interpretation. The detector complements other assessments and does not independently decide water safety or purification.

7.5 Limitations and threats to validity

Threats include similar morphology between taxa, preparation or staining artefacts, blur, overlapping cells, limited reference specimens, class imbalance and domain shift. A model may learn slide background or laboratory batch characteristics rather than biological identity. Keeping related data in one split and holding out acquisition groups reduces, but does not eliminate, this risk.

Sample-level identification may not resolve individual objects in mixed fields. Public datasets may provide inadequate provenance, too few independent isolates or no box labels. Many cells from one slide do not establish broad biological diversity. Unknown confidence behaviour can remain unreliable despite threshold tuning.

Detection boxes do not establish viability, toxin production or pathogenic strain identity. Image counts require additional sampling calibration before any concentration inference, and negative images cannot exclude organisms outside the imaged area or below the detection capability. The final thesis must preserve these limits.

7.6 Implications

If validated, the component could help reviewers inspect microscopy fields and prioritize uncertain observations for further assessment. Potential users include research laboratories, environmental monitoring teams and educational projects. No measured public-health benefit, cost saving or deployment readiness is established.

Sustaining the system would require reference-data maintenance, expert review, model monitoring and clear responsibility for errors. Commercial use would require separate assessment of user demand, licences, laboratory workflows and applicable requirements. The prototype should not be marketed as an unrestricted species-identification or water-certification tool.

8. CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

This thesis draft specifies a Microorganism (Bacteria) Detection component using CNN-based object detection. It outlines data curation, instance annotation, a YOLO/Faster R-CNN comparison, taxonomic acceptance rules and integration with the reporting service.

The central design requirement is to match the specificity of the output to the strength of reference and validation evidence. Species, genus and unidentified results must remain distinct. The component's performance has not yet been demonstrated; the next stage is to obtain suitable data, implement the pipeline and execute the planned evaluation.

8.2 Achievement of objectives

Table 8.1 distinguishes completed draft specification from pending research and development work. No objective requiring a dataset, trained model or measured result is represented as achieved.

Table 6: Objective status

Objective	Draft specification	Remaining evidence
O1	Imaging/label/reference policy	Confirmed collaborators, modality and target taxa
O2	Dataset and split design	Curated instance-labelled microscopy and manifests
O3	Model comparison protocol	Trained checkpoints and held-out predictions
O4	Taxonomic fallback/rejection design	Independent evidence of accuracy and coverage
O5	Output contract and testing plan	Implemented service, integration and measurements

8.3 Individual research and product contributions

The individual contribution presently documented is research framing, label hierarchy, reference-data requirements, system design and evaluation plan. Planned implementation contributions include annotated-data processing, training workflows, detection and rejection logic, an inference API and integration tests.

The final thesis should record actual work with code revisions, reviewed annotations, experiment logs and team acknowledgements. This draft does not substitute for those artefacts.

8.4 Future directions

Immediate steps are to identify a qualified laboratory collaborator or suitable reference dataset, settle the imaging modality, define a small supported taxonomic panel, and complete a reviewed annotation pilot. Test whether species discrimination is plausible before scaling the dataset. Implement and evaluate localization and identity as separate outcomes.

Later work could explore segmentation of touching cells, additional microscopy modalities, richer unknown detection, multi-laboratory evaluation and prospectively collected water samples. Concentration or viability assessment would require a separately specified and validated study. New taxa should enter the supported output set only after adequate reference data and evaluation.

REFERENCES

[1] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," *arXiv preprint arXiv:1506.02640*, 2015. [Online]. Available: https://arxiv.org/abs/1506.02640

[2] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *arXiv preprint arXiv:1506.01497*, 2015. [Online]. Available: https://arxiv.org/abs/1506.01497

[3] B. Zieliński, A. Plichta, K. Misztal, P. Spurek, M. Brzychczy-Włoch, and D. Ochońska, "Deep learning approach to bacterial colony classification," *PLOS ONE*, vol. 12, no. 9, Art. no. e0184554, 2017, doi: 10.1371/journal.pone.0184554.

[4] C. Spahn *et al.*, "DeepBacs for multi-task bacterial image analysis using open-source deep learning approaches," *Communications Biology*, vol. 5, Art. no. 688, 2022, doi: 10.1038/s42003-022-03634-z.

[5] G. Kim *et al.*, "Rapid species identification of pathogenic bacteria from a minute quantity exploiting three-dimensional quantitative phase imaging and artificial neural network," *Light: Science & Applications*, vol. 11, Art. no. 190, 2022, doi: 10.1038/s41377-022-00881-x.

[6] E. Moen, D. Bannon, T. Kudo, W. Graf, M. Covert, and D. Van Valen, "Deep learning for cellular image analysis," *Nature Methods*, vol. 16, no. 12, pp. 1233–1246, 2019, doi: 10.1038/s41592-019-0403-1. Nature Methods

[7] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Medical Image Computing and Computer-Assisted Intervention—MICCAI 2015*, Lecture Notes in Computer Science, vol. 9351. Cham, Switzerland: Springer, 2015, pp. 234–241, doi: 10.1007/978-3-319-24574-4_28. lmb.informatik.uni-freiburg.de

[8] C. Stringer, T. Wang, M. Michaelos, and M. Pachitariu, "Cellpose: A generalist algorithm for cellular segmentation," *Nature Methods*, vol. 18, no. 1, pp. 100–106, 2021, doi: 10.1038/s41592-020-01018-x. Nature Methods

[9] J.-B. Lugagne, H. Lin, and M. J. Dunlop, "DeLTA: Automated cell segmentation, tracking, and lineage reconstruction using deep learning," *PLOS Computational Biology*, vol. 16, no. 4, Art. no. e1007673, 2020, doi: 10.1371/journal.pcbi.1007673. PLOS Computational Biology

[10] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, “On calibration of modern neural networks,” in *Proc. 34th Int. Conf. Machine Learning*, vol. 70, 2017, pp. 1321–1330. [Online]. Available: https://proceedings.mlr.press/v70/guo17a.html.

GLOSSARY

Annotation: The assignment of reference labels and spatial boundaries to objects in microscopy images for model training and evaluation.

Average precision (AP): A measure summarizing the precision–recall relationship for a detection class under specified matching criteria.

Bounding box: A rectangle indicating the location and approximate extent of a microorganism or defined object within an image.

Confidence calibration: The assessment or adjustment of prediction scores so that they more closely reflect observed prediction correctness.

Confidence score: A model-generated score indicating support for prediction. It does not independently confirm organism identity or pathogenicity.

Convolutional neural network (CNN): A neural network that uses convolutional operations to learn spatial features from images.

Coverage: The proportion of eligible cases receiving an accepted prediction under a specified reporting policy.

Data augmentation: Controlled transformations of training images that increase variation while preserving valid reference annotations.

Data leakage: The unintended sharing of related information between training, validation and test partitions, potentially inflating measured performance.

Domain shift: A difference in specimen characteristics or imaging conditions between model development and subsequent evaluation or use.

False negative: A reference target that the model fails to detect under the specified evaluation criteria.

False positive: A predicted detection that does not correctly match a reference target under the specified evaluation criteria.

Faster R-CNN: A two-stage object-detection framework that generates candidate regions before classifying objects and refining their bounding boxes.

Field of view: The specimen area captured in a single microscope image.

Genus: A biological classification ranks above species that groups related species.

Ground truth: Reviewed reference annotations used for training and evaluation, subject to the limitations of the reference method and annotation process.

Inference: The application of a trained model to new microscopy images to generate predictions.

Instance annotation: A spatial boundary and reference label assigned to an individual cell, organism or explicitly defined object.

Instance segmentation: The identification of individual objects and their pixel-level boundaries within an image.

Intersection over union (IoU): The intersection area of predicted and reference bounding boxes divided by their union area.

Mean average precision (mAP): Average precision aggregated across classes and, where specified, multiple IoU thresholds.

Microscopy modality: The imaging technique is used to observe a specimen, such as bright-field, phase-contrast or fluorescence microscopy.

Morphology: The observable shape, size and structural characteristics of an organism, which may not uniquely establish its taxonomic identity.

Non-maximum suppression (NMS): A post-processing method that reduces redundant overlapping detections using prediction scores and an overlap threshold.

Open-set rejection: Withholding a known-class assignment when an input is assessed as unfamiliar or unsupported by the model's evaluated categories.

Precision: The proportion of predicted detections that correctly match reference targets under the specified evaluation criteria.

Recall: The proportion of reference targets correctly detected under the specified evaluation criteria.

Reference identification: An independently supported organism identity used to establish training or evaluation labels.

Species: A biological classification rank below genus, used in this study only where reference evidence and validation support that level of identification.

Specimen-wise splitting: Dataset partitioning that keeps related images, fields, tiles and derivatives from the same specimen within one partition.

Taxon: A named biological classification group, such as a genius or species.

Taxonomic rank: The hierarchical level associated with an organism label, such as genus or species.

Tiling: Dividing a large microscopy image into smaller regions for processing, followed by coordinating restoration and reconciliation of overlapping detections.

Transfer learning: Adapting a model pretrained on a source dataset to a target task or dataset.

Unknown or unidentified outcome: A result that retains a detected object's location without assigning an insufficiently supported taxonomic identity.

YOLO (You Only Look Once): A family of single-stage object detectors that predict object locations and categories through a unified detection network.

APPENDICES

Appendix A: Microscopy Data Collection and Annotation Instruments

This appendix presents the proposed instruments for preparing the microorganism-detection dataset. These materials do not represent completed data collection or laboratory validation.

Table 7: Image Acquisition Record

Field	**Information to record**
Sample and specimen identifiers	Unique codes linking the water sample to its microscopy preparation
Slide and image identifiers	Unique codes for each slide and captured field
Acquisition details	Date, session, microscope and camera
Imaging modality	Bright-field, phase-contrast, fluorescence or other documented technique
Optical information	Objective magnification, numerical aperture where available and pixel calibration
Preparation information	Documented preparation or staining method
Image properties	Dimensions, file format, focus and exposure quality
Reference identification	Independently supported identity, method and evidence source
Data provenance	Image ownership, permission and applicable licence

A.2 Instance Annotation Record

Each annotated instance will include its image identifier, bounding-box coordinates, reference label, taxonomic rank, reference-identification basis and reviewer identifier. Additional fields will record overlaps, occlusion, uncertainty and unresolved clusters.

A sample-level identification will not automatically be assigned to every visible object in a mixed microscopy field. Where instance identity cannot be supported, the annotation will retain an unidentified label or follow the documented exclusion policy.

A.3 Annotation Review Procedure

Two reviewers will independently annotate a pilot subset. Agreement on object boundaries and reference labels will be assessed before disagreements are adjudicated. The annotation guide will define procedures for touching cells, partial objects, unresolved clusters, preparation artefacts and uncertain identities. Revisions and exclusion decisions will be retained in an audit record.

Appendix B: Taxonomic Reporting and Evaluation Protocol

Table 8: Proposed Reporting Rules

Evidence condition	**Permitted output**
Species identity supported by reference data and class-specific validation	Species prediction with confidence and identification limitations
Species assignment unsupported, but a separately evaluated genus decision is supported	Genus prediction with explicit rank
Object detected without sufficient taxonomic evidence	Unidentified object with its bounding box
Dense cluster cannot be resolved into individual cells	Cluster or unresolved-region output according to the annotation policy
Image quality or modality falls outside the supported domain	Rejected or limited-result status with a reason
No retained detections in an eligible image	No-detection status without a water-safety conclusion

B.2 Experimental Record

Each experiment will document the dataset and annotation versions, independent specimen counts, split manifest, model architecture, pretrained checkpoint, software versions, hyperparameters, random seed and hardware.

Evaluations will report localization and taxonomic identification separately. Required measures include class-wise average precision, precision, recall, small-object performance, identification errors, unknown-target false acceptance, known-target false rejection, prediction coverage and inference latency. Results will be stratified by relevant acquisition conditions and object characteristics.

Appendix C: Proposed Software and Integration Tests

The following cases specify planned checks. Numerical tolerance and acceptance criteria will be established before execution.

Test ID	**Test condition**	**Expected behaviour or assessment**
01	Valid supported microscopy image	Return a structured response with the original sample and image identifiers.
02	Corrupt or unsupported file	Reject the input with an explicit error status.
03	Missing required acquisition metadata	Return a validation error or documented limited-result status.
04	Resizing or padding applied	Restore bounding boxes to original-image coordinates within the specified tolerance.
05	The same object appears in overlapping tiles	Reconcile duplicate detections before reporting or counting.
06	Two neighbouring cells are distinguishable	Assess whether duplicate suppression incorrectly removes a genuine instance.
07	Unsupported species assignment	Return an unidentified result or a separately supported genus prediction.
08	Unfamiliar organism or preparation artefact	Measure false acceptance and rejection performance on independent examples.
09	Unresolved dense cluster	Apply the cluster policy without inventing individual-cell counts.
10	Eligible image with no retained detections	Return a no-detection status without implying absence of microorganisms from the sample.
11	Inference timeout or service failure	Return an explicit failure status with traceable identifiers.
12	Results transferred to the reporting component	Preserve taxonomic rank, confidence, limitations, model version and image-level count meaning.

For every executed test, record the input, configuration, expected outcome, observed outcome, execution date and supporting evidence. Pass/fail decisions will apply only where predefined acceptance criteria exist.

Appendix D: Academic Integrity and AI Use Disclosure

Generative AI assisted with drafting, restructuring and language refinement of thesis content, identifying candidate references and formatting citations. Assisted sections include the abstract, contextual descriptions, glossary and appendix materials.

The candidate remains responsible for verifying source publications, checking the accuracy of all statements and ensuring that the thesis reflects the research conducted. AI-generated text does not constitute laboratory evidence, reference identification or experimental validation. No generated performance values are presented as measured findings. The final disclosure will identify the tools and assistance used and follow the applicable institutional requirements.